Stan Fris

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Education

University of Amsterdam – Master Artificial Intelligence *September 2024 - June 2026*

- GPA 8.4/10
- Thesis: *Feature-Based Propensity estimation for Unbiased Learning to Rank* (9.0), Supervisor: Dr. Harrie Oosterhuis

University of Amsterdam – Bachelor Artificial Intelligence *September 2021 - June 2024*

- GPA 8.7/10, *Cum Laude*
- Thesis: *Predictive Railway maintenance planning* (8.5), Supervisor: Dr. Gregor Behnke

Experience

Visiting Research Intern, National Institute of Informatics (NII) - Tokyo, Japan *August 2026 – Present*

- Developing a benchmark for scientific paper re-finding using multimodal and textual document representations.
- Building LLM-assisted dataset construction and evaluation pipelines for scientific information retrieval.

Research Intern, University of Amsterdam – Amsterdam *August 2025 – January 2026*

- Developed a practical framework for diagnosing identifiability in neural two-tower click models using JAX/Flax.
- Evaluated the method on synthetic and real-world click data, analyzing when relevance and position-bias parameters can be reliably identified. Resulted in a full paper accepted at SIGIR 2026.

Machine Learning Intern, ProRail – Utrecht *October 2023 – March 2024*

- Developed a predictive maintenance-planning system for the Dutch railway network using time-series forecasting and optimization.
- Combined LSTM-based predictions with deterministic Max-SAT planning to support maintenance scheduling.

Teaching-Assistant and Tutor, University of Amsterdam – Amsterdam *August 2023 – July 2026*

- Teaching-Assistant in the Bachelor Artificial Intelligence, teaching courses on topics including machine learning, Calculus and Natural Language Processing
- Independently taught tutorials and seminars for classes of approximately 25 students.
- Organized courses working with companies, guided student-stakeholder contact.

Publications

Feature-based Propensities for Counterfactual Learning to Rank *To be Released*
Stan Fris, David Vos, Harrie Oosterhuis
ACM SIGIR Conference on Information Retrieval in the Asia Pacific region (SIGIR-AP), 2026

Diagnosing Identifiability in Two-Tower Models for Unbiased Learning to Rank [Paper][Code]
Stan Fris, Philipp Hager
ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2026

Improving the Efficiency and Effectiveness of LLM Knowledge Distillation for Conversational Search [Paper][Code]
Stan Fris, Jan Hutter, Jan-Henrik Bertrand, Simon Lupart, Mohammad Aliannejadi
The SCAI Workshop at the 49th ACM SIGIR Conference (non-archival), 2026

A Systematic Reproducibility Study of BSARec for Sequential Recommendation [Paper][Code]
Jan Hutter*, Hua Chang Bakker*, **Stan Fris***, Madelon Bernardy*, Yuanna Liu
European Conference on Information Retrieval (ECIR), 2026, *Equal Contribution

Skills

Programming: Python, C, C++, SQL
ML Frameworks: PyTorch, JAX, Flax (NNX), Hugging Face
Systems & Tools: Slurm, Linux, Weights & Biases, Git, Hydra
Languages: Dutch: Native, English: Fluent

Selected Machine Learning Projects

Fairness of Vision Language models in Healthcare (Pytorch) [Paper][Code]

- Conducted a reproducibility study of a framework for improving group fairness of CLIP.
- Expanded the original framework to be able to account for multiple types of group fairness at the same time, improving the flexibility and fairness of the model.

Domain Adaptation for Low-Resource Biomedical Translation (Pytorch) [Paper][Code]

- Worked on LLM finetuning for domain adaptation of neural translation models in low resource settings
- Applied fine-tuning techniques such as LoRA to effectively adapt LLMs to new tasks.

Open Source

Multimodal Tip-of-The-Tongue Retrieval for Scientific Documents [Hugging Face][Code]

- Developed and open-sourced a scientific retrieval benchmark covering 100K documents across five research domains.
- Built LLM-based pipelines for document processing, multimodal query generation, and automatic quality evaluation.
- Implemented and evaluated lexical, dense, and multimodal retrieval baselines, with reproducible experiment configurations.

Grants & Awards

SIGIR Student Travel grant *2026*

ELLIS & Qualcomm Student Travel grant *2026*

MSc Artificial Intelligence Student Travel Grant *2026*

Activities

Programme Committee, ECIR 2026

Education Committee Artificial Intelligence – Amsterdam *October 2023 – July 2026*

- Represent student interests in programme-level discussions on curriculum and education quality
- Communication with Programme Director, overseeing education quality and consistency.

Faculty Student Council UvA, Amsterdam- Chair – Amsterdam *September 2022 – September 2023*

- Chaired the council and represented students in discussions with faculty leadership and university governance bodies.
- Worked on files including diversity, inclusivity, and events.